

Goiter, multinodular 1, with or without Sertoli-Leydig cell tumors

Pleuropulmonary blastoma

Rhabdomyosarcoma, embryonal, 2

GLOW syndrome, somatic mosaicism

Dicer 1, ribonuclease III

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HS MKSPALQPLSMAGLQLMTPASSPMGPFPGFLPWQQEAIHNDNIYTPRKYQVELLEAALDHNITVCLNTGSGKTFIAVLLTKELSYQIRGDFS
MM MKSPALQPLSMAGLQLMTPASSPMGPFPGFLPWQQEAIHNDNIYTPRKYQVELLEAALDHNITVCLNTGSGKTFIAVLLTKELAHQIRGDLN
RN MKSPALQPLSMAGLQLMTPASSPMGPFPGFLPWQQEAIHNDNIYTPRKYQVELLEAALDHNITVCLNTGSGKTFIAVLLTKELAHQIRGDLN
XT -----MAGLQLMTPASSPMGPFPGFLPWQQEAIHNDNIYTPRKYQVELLEAALDHNITVCLNTGSGKTFIAVLLTKELSYQIRGDFS
DR -----MAGLQLMTPASSPMGPFPGFLPWQQEAIHNDNIYTPRKYQVELLEAALDHNITVCLNTGSGKTFIAVLLTKELSHQIRGE--
CE -----MVRVRADLQCFNPRDYPQVELLDKATKKNTIVQLGTGSGKTFIAVLLTKELSYGVQLFAPLD
DM -----MAFHWCNDNLHTTPTPRDFQVELLATAYERNITICLGHRSSKEFIALKLQELSSRRARR--
                                     10         20         30         40         50         60         70         80         90

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HS RNGKRTVFLVNSANQVAQQVSAVRTHSDLVKVGSEYSLNEVNASWTKER----WNQEFTKHQVLIIMTCYVALNVLNKNGYLSLSDINLLVFD
MM PHAKRTVFLVNSANQVAQQVSAVRTHSDLVKVGSEYSDLEVNASWTKER----WSQEFTKHQVLIIMTCYVALNVLNKNGYLSLSDINLLVFD
RN PHAKRTVFLVNSANQVAQQVSAVRTHSDLVKVGSEYSLNEVNASWTKER----WSQEFTKHQVLIIMTCYVALNVLNKNGYLSLSDINLLVFD
XT KNTKRTVFLVNSEKQVSQQVSAVRTHDVLKVGSEYSDQKTCQWAKER----WYLEFETHQVLMVMTCHIFLNVLKSNGVLSLSDINLLVFD
DR -NGGKRTVFLVNAASSVAQQASTVHSDLVKVGQYMSD-MTSWPPEEM----WNQREMIENQVLIIMTCIFLHLVHNKNGYLPKSLINLLVFD
CE GQKKRAFLVVEKVNVLVEQQAIRHIEVHTSFVKYGVHQGTSSGLWDSKE----QDQFMKRHHVMTITACQLLDLRHAYLIEDMCVLIFD
DM -HGRVSVLSCVEGVTSTEPSCSIYTM-----THLTDLRVWQEQPDMQIPFDHCWTDYHVSILRPEGFLYLLLETRELLSSVELTVLE
                                     100        110        120        130        140        150        160        170        180

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HS ECHLAIL-DHPYREIMK-----LCENCPSCPRILGLTASILNGKCDPEELEEIKQLEKILKLSNAETATDLVLVDRYTSQPCEIVDVGCPFT
MM ECHLAIL-DHPYREIMK-----LCESCPSCPRILGLTASILNGKCDPEELEEIKQLERILRSDAETATDLVLVDRYTSQPCEIVDVGCPFT
RN ECHLAIL-DHPYREIMK-----LCDSCPSCPRILGLTASILNGKCDPDELEEIKQLEKILKLSGAETATDLVLVDRYTSQPCEIVDVGCPFT
XT ECHLAIQ-DHPYREIMK-----ICESCQPCPRILGLTASILNGKCDPRDLEEIKQLEELILRSNAETATDLVLVDRYASQPCEIVDGGPPYI
DR ECHLAIT-GHPYREIMK-----ICEGQPCSPRILGLTASILNGKCDPCDLEEIKQNLKELILQSSAETATDLVLVDRYASQPREELVDCGQYQ
CE ECHHALGSGQHPYRISIMVDYHLKKDKPVPRLVGLTASLIKAVAPEKLEMLQKKLESAMDVSIEATSDLVLSKYGAKPYEVVITICKDFE
DM DCHDSAVY-QRIRPLFENHIMPAPPADRPRLGLAGPLHSAAGLEQLLSAMLATLEQSVLCQIETASDITVTVLYCSRPHHEYIVQCAPFE
                                     190        200        210        220        230        240        250        260        270

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Efs (RCV000023523)



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HS DRSG---LYERLLMELEELNFINDCN-----ISV--HSKERDSTLISKQILSDCRAVLVVLGPWCADKVGAGMMVRELOKYIKHE
MM DRSG---LYERLLMELEELNFINDCN-----VAV--HSKERDSTLISKQILSDCRAVLVVLGPWCADKVGAGMMVRELOKYIKHE
RN DRSG---LYERLLMELEELNFINDCN-----VSV--HSKERDSTLISKQILSDCRAVLVVLGPWCADKVGAGMMVRELOKYIKHE
XT DSG---LYQRLLNELDEALNFINDCN-----IST--HSKERDSTLISKQILSDCQVTLVVLGPWCADKVGAGMMVRELOKYIKHE
DR DQSG---LERILLNELDEALNFINDCN-----LSV--HREDRPTFISKQVINDCRAVLTVLGPWCADKAGIMVRELOKYIKHE
CE IGCLGIPNFDVTWIEFDVTAFAVNTTT-----EFH--PDLDLPRRPKDSLKTTRAVFQLGPWAAWRTAQVWEKELGKILKSQ
DM MDELSLVADVYN---THKSFLLDHRYPDYEIYGTDQFMDELKIDIPPKVDPLNVINSLLVVLHEMGPWCTQRAAHFYQCNEKLKVKT
                                     280        290        300        310        320        330        340        350        360

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HS QEE-LHRKFLLFTDTFLRKIHALCEEHFS-----PASLDLKFTVPKVIKLLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
MM QEE-LHRKFLLFTDTLLRKIHALCEEHFS-----PASLDLKYYTPKVMKLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
RN QEE-LHRKFLLFTDTLLRKIHALCEEHFS-----PASLDLKYYTPKVMKLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
XT QEE-LHRKFLLFTDTLLRKIHALCEEHFS-----PASLDMKFVTPKVIKLLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
DR QEE-LNRKFLLFTDTLLRKIHALCEEHFS-----PASLDLKFTVPKIRLLEILHEYPKFERQQFESV-EWYN-NRNQDNVYSWSDS
CE VLPDKTRFLNMAKTSMITIKRLLEPEMKTKI--SIEALRYPQVIRLFEILETFNPEFQKE-----KTGATLMKSTALINRYCKSLPSDIFTRLVPHNQII
DM -PHERHYLLYCLVSTALIQLYSLCEHAHHRHLGSGSDSRQTIERYSSPKVYRLLQTLRCFKPEEVHTQADGLRMRMRHQVDQADFNRLLSHT
                                     370        380        390        400        410        420        430        440        450

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HS -----EDD-----DEDEE-----IEEKE-----KPETNFPSPTNILCG
MM -----EDD-----DDDEE-----IEEKE-----KPETNFPSPTNILCG
RN -----EDD-----DDDEE-----IEEKE-----KPETNFPSPTNILCG
XT -----EDD-----DEDEE-----IEEKE-----KTETSFPSPTNILCG
DR -----EDD-----DEDEE-----AAKE-----KTEANFPSPTNILCG
CE -----EDD-----DEDEE-----AAKE-----KTEANFPSPTNILCG
DM LESKCRMVDQMQPPPTETRALVATLEQILHTTEDRQTNRSAARVTPPTTPAHAKPKPSSGANTAQPRTRRRRVYTRRHRHDHNDGSDTLCA
                                     460        470        480        490        500        510        520        530        540

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E>X (RCV000004726)

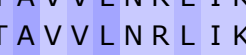


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HS IIFVERRYTAVVLNRLIKEAGKQDPELAYISSNFITGHGIGKNQPRNKQMEAERFKQEEVLKRFRAHETNLLIATSVIEEGVDIPKCNLV
MM IIFVERRYTAVVLNRLIKEAGKQDPELAYISSNFITGHGIGKNQPRNKQMEAERFKQEEVLKRFRAHETNLLIATSVVEEGVDIPKCNLV
RN IIFVERRYTAVVLNRLIKEAGKQDPELAYISSNFITGHGIGKNQPRNKQMEAERFKQEEVLKRFRAHETNLLIATSVVEEGVDIPKCNLV
XT IIFVERRYTAVVLNRLIKEAGKQDPELAYISSNFITGHGIGKNQPRNKQMEVEFRKQEEVLKRFRAHETNLLIATSVIEEGVDIPKCNLV
DR IIFVERRYTAVVLNRLIKEAGKQDPELAYISSNFITGHSIGKNQPRNKQMEVEFRKQEEVLKRFRAHETNLLIATSVIEEGVDIPKCNLV
CE IIFVDQRYIAYSLLMMMRHISKEWPKFKFVNPDYVVGASGRNLAS--SDSQGLHKRQTEVLRRFRHNEINCLATISVLEEGBVDPKCNLV
DM LIYCNQNHITARVLFELLAEISRRDPDLKFLRCQYITDRVAD-PTTEPKAELEHRRQEEVLKRFMRHDCNVLIGTSVLEEGBVDPKCNLV
                                     550        560        570        580        590        600        610        620        630

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R>X (RCV000004729)



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HS VRFDLPTEYRSYVQSKGRARAPISNYIMLADTDKIK-----PASLDLKFTVPKVIKLLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
MM VRFDLPTEYRSYVQSKGRARAPISNYIMLADTDKIK-----PASLDLKYYTPKVMKLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
RN VRFDLPTEYRSYVQSKGRARAPISNYIMLADTDKIK-----PASLDMKFVTPKVIKLLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
XT VRFDLPSEYRSYVQSKGRARAPISNYIMLADSDKIK-----PASLDMKFVTPKVIKLLEILRKYPYERQQFESV-EWYN-NRNQDNVYSWSDS
DR VRFDLPTEYRSYVQSKGRARAPISNYIMLADSERTK-----PASLDMKFVTPKIRLLEILHEYPKFERQQFESV-EWYN-NRNQDNVYSWSDS
CE IKFDRPLDMRSYVQSKGRARAPISNYIMLADSERTK-----PASLDMKFVTPKIRLLEILHEYPKFERQQFESV-EWYN-NRNQDNVYSWSDS
DM VRWDPTTTYRYSYVQCKGRARAAPAYHVLVAPSYKSPTVGSVQLTDRSHRYICATGDTTEADSDSDSAMPNSSGSDPYTFGTARGTVKI
                                     640        650        660        670        680        690        700        710        720

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HS -----SF-----
MM -----SF-----
RN -----SF-----
XT -----AF-----
DR -----TF-----
CE -----AC-----
DM LNPEVFSKQPPTACDIKLQEIQDELPAQAQLDTSNSSDEAVMSNTSPSPSSSTEQKSRRFQCELSSLTEPEDTSDTTAEIDTAHSLASTT
                                     730        740        750        760        770        780        790        800        810

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HS ---EEDLKTYKAEIKILRNKCSKSVDTGET--DIDPVMDDDDVFPYPVLR-PDDGGPRVTINTAIGHINRYCARLPSDPFTHLAPKCRTR
MM ---EEDLKTYKAEIKILRNKCSKSDGAEA--DVHAGVDDDEDAFPYPVLR-PDDGGPRVTINTAIGHINRYCARLPSDPFTHLAPKCRTR
RN ---EEDLKTYKAEIKILRNKCSKSDGAEA--DVHAGVDDDDVFPYPVLR-PDDGGPRVTINTAIGHINRYCARLPSDPFTHLAPKCRTR
XT ---EEDLKTYKAEIKILRNKCSKSDCGNT--ESEPVDDEIDFPYPVLR-QDDGSPRVTINTAIGHINRYCARLPSDPFTHLAPKCRTR
DR ---QEDLKTYKAEIKILRNKCSKAECNDF--ELEPVTDDDNVLPYPVLR-SEDGGPRVTINTAIGHINRYCARLPSDPFTHLAPKCRTR
CE ---DSDLDKDFQIIEKILLSRHRTVNPPIEDSDRFEEDVDQMEPFYVVE--KTGATLMKSTALINRYCKSLPSDIFTRLVPHNQII
DM KDLVHQAQYREIEQMLLSKCANTEPPEQEQSEAERFSA--CLAAAYRKPHELLTGASVDLGSAIALVNYKYCARLPSDTFTKLTALWRCT
                                     820        830        840        850        860        870        880        890        900

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HS ELPD---GTFYSTLYLPINSLRASIV-GPPMSCVRLAERVVALICCEKLHKIGELDDHLMVPVGKETVKYEEE-LD-L---HD---EEET
MM ELPD---GTFYSTLYLPINSLRASIV-GPPMDSVRLAERVVALICCEKLHKIGELDEHLMVPVGKETVKYEEE-LD-L---HD---EEET
RN ELPD---GTFYSTLYLPINSLRASIV-GPPMGCVRLAERVVALICCEKLHKIGELDEHLMVPVGKETVKYEEE-LD-L---HD---EEET
XT ELPD---GLYRSTLYLPINSLRASIV-GPPMNGCLADRAVALICCKLHEIGELDDHLMVPVGKETVKYEEE-LD-L---HD---EEET
DR EMNT---GGYRSTLYLPINSLRPVPT-GPVMNCARLAEKAVALLCCEKLHKIGELDDHLMVPVGKETVKYEEE-LD-L---HD---EEET
CE PIEENGVTYKCAELLPINSPIKHAIVLNKPMPNKKTQMAVALEACRQLHLEGELEDNLLPKGRESIAKLEHIDEE--PD---EYAP
DM RNERAGVTLFQYTLRLPINSPKLHIV-GLPMPTQTLARRLAALQACVLEHRIIGELDDQLPAGKEGRFALPDWECFELEPEDEQIVQL
                                     910        920        930        940        950        960        970        980        990

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Tfs (RCV000240963)



IYdel PAZ (RCV001398192)

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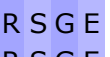
HS SVPGRPGSTKRRQCYPKAIPCELRDSYPRPDQPCYLYVIGMVLTTPLPDELNFRRRKLYPPEDTTRCFGILTAKPIQIPHFPVYTRSGE
MM SVPGRPGSTKRRQCYPKAIPCELRDSYPRPDQPCYLYVIGMVLTTPLPDELNFRRRKLYPPEDTTRCFGILTAKPIQIPHFPVYTRSGE
RN SVPGRPGSTKRRQCYPKAIPCELRDSYPRPDQPCYLYVIGMVLTTPLPDELNFRRRKLYPPEDTTRCFGILTAKPIQIPHFPVYTRSGE
XT SVPGRPGSTKRRQCYPKAIPCELRDSYPRPDQPCYLYVIGMVLTTPLPDELNFRRRKLYPPEDTTRCFGILTAKPIQIPHFPVYTRSGE
DR SVPGRPGSTKRRQCSKPAIPCELRDSYPRPDQPCYLYVIGMVLTTPLPDELNFRRRKLYPPEDTTRCFGILTAKPIQIPHFPVYTRSGE
CE GIAAKVGSKKRQLYDKKIAALNESFVEADKECFIYAFELERFRAEALTLPKRRKFEDPFNYEYCFGLSALKEPKIPFPFVFLRQGN
DM SDEPRPGTTRKQYYYKRIASEFCDCRPVAGAPCYLYFIQLTLQCPIPEEQNTRGRKIYPPEDAQQGFILTTKRIPLKLSAFSIFTRSGE
                                     1000       1010       1020       1030       1040       1050       1060       1070       1080

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S>F (RCV000023526)



EX18del PAZ (RCV001201069)



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HS VTISIELKKSGFMLSQMLLELITRLHQYIFSHILRLEKP-ALEFKPTG-ADSAYCVLPNVVNDSSGTLIDDFKFMEDIEKS-EARIGIPS
MM VTISIELKKSGFMLSQMLLELITRLHQYIFSHILRLEKP-ALEFKPTG-AESAYCVLPNVVNDSSGTLIDDFKFMEDIEKS-EARIGIPS
RN VTISIELKKSGFMLSQMLLELITRLHQYIFSHILRLEKP-ALEFKPAG-AESAYCVLPNVVNDSSGTLIDDFKFMEDIEKS-EARIGIPS
XT VTISIELKKSGFMLSQMLLELITRLHQYIFSHILRLEKP-ALEFKPTV-ADCAVCVLPNVVNDSSGTLIDDFKFMEDIEKS-EARTGIPN
DR VTISIELKKSGFMLSQMLLELITRLHQYIFSHILRLEKP-ALEFKPVE-ADSAYCVLPNVVNDSSGTLIDDFKFMEDIEKS-EARIGIPN
CE MKVRLIVAPKKTITVAEQLEIQLFHNHLYFTQVLQCKTGNLEFDGTSNAPLNTLIVPLNKRKDDMSYTIMMKYVSEVAVANMENMPRIK
DM VKVSLLELAKERVILTSEQIVCINGFLNYTFTNVLLQKF-LMLFDPDSTENCVFIPT--VKAPAGGKHIDQWFLLELIQANGNTMPRAVP
                                     1090       1100       1110       1120       1130       1140       1150       1160       1170

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R>X (RCV000004727)



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HS TKYTKETPFVFKLEDYQDAVIIPR---YRNFDPQHRFYVADVYDTLTPLSKFPSPPEYETFAEYYKTKYNLDLTNLNQPLLDVDHTSSRLN
MM TKYTKETPFVFKLEDYQDAVIIPR---YRNFDPQHRFYVADVYDTLTPLSKFPSPPEYETFAEYYKTKYNLDLTNLNQPLLDVDHTSSRLN
RN TKYTKETPFVFKLEDYQDAVIIPR---YRNFDPQHRFYVADVYDTLTPLSKFPSPPEYETFAEYYKTKYNLDLTNLNQPLLDVDHTSSRLN
XT TQYSAESPFIKLEDYQDAVIIPRQVIYRNFDPQHRFYVADVYDTLTPLSKFPSPPEYETFAEYYKTKYNLDLTNLNQPLLDVDHTSSRLN
DR TQYTKQNPFIFKLEDYQDAVIIPR---YRNFDPQHRFYVADVYDTLTPLSKFPSPPEYETFAEYYKTKYNLDLTNLNQPLLDVDHTSSRLN
CE D--EVRRQYKFNAEDYKDAIVMPW---YRNLEQPVFYVAAELPEWRVSPSKFPDTHFETTFNEFYTKYKYLEITYDQNSQLDVPDFTSTRLN
DM D--EERQAQPFDPQRFQDAVMPW---YRNDQDPQYFYVAAICPHLSPLSCFPGDNYRTFKHYLYVKYGLTQNTSQPLLDVDHTSARLN
                                     1180       1190       1200       1210       1220       1230       1240       1250       1260

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HS LLTPRHLNQKGKALPLSSA-----EKR---KAKWESLQNKQILVPELCAIHPIPASLWRKAVCLPSILYRLHCLLTAEEELRAQTA
MM LLTPRHLNQKGKALPLSSA-----EKR---KAKWESLQNKQILVPELCAIHPIPASLWRKAVCLPSILYRLHCLLTAEEELRAQTA
RN LLTPRHLNQKGKALPLSSA-----EKR---KAKWESLQNKQILVPELCAIHPIPASLWRKAVCLPSILYRLHCLLTAEEELRAQTA
XT LLTPRHLNQKGKALPLSSA-----EKR---KAKWESLQNKQILVPELCAIHPVPASLWRKAVCLPSILYRLHCLLTAEEELRAQTA
DR LLTPRHLNQKGKALPLSSA-----EKR---KAKWESLQNKQILVPELCAIHPIPASLWRKAVCLPSILYRLHCLLTAEEELRSQTA
CE LLQPRIQNQPRRSRTVSNSSSTSNIPQASASDSKESNTSVPHSSQRQILVPELMDIHPISATLWNVIAALPSIFYRVNQQLLTDELRETI
DM FLTPRYVNRKGVALPTSSEET-----KRAKRENLEQKQILVPELCTVHPFPASLWRKAVCLPSILYRLHCLLTAEEELRAQTA
                                     1270       1280       1290       1300       1310       1320       1330       1340       1350

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HS SDA-GVGVRSLPADFRYPNLDFGWKK---SID--SKSFISI-----SNSSAEND-----NY-----CKHST-
MM SDA-GVGVRSLPADFRYPNLDFGWKK---SID--SKSFIST-----CNSSLAESD-----NY-----CKHSTT
RN SDA-GVGVRSLPADFRYPNLDFGWKK---SID--SKSFIST-----CNSSLAESD-----NY-----CKHSTT
XT IDA-GVGKSLPADFRYPNLDFGWKR---SID--SKTFISN-----QSSSVSE-----SD-----CRNLKT
DR IDA-GVGAKT-LPPDFRYPNLDFGWKK---SID--SKSFISC-----PSACMEEDD-----DH-----CKLGS
CE VKAFGEKVT-LDDNVWNSLAYATEYEEKQII--VKIKQILRD--LNQKSIEDQE-----RE-----TRENDK
DM ADL-GLGRQQIEDDEFEPMLDFGWSLSEVLKKSRESKQKESLKDDTINKGDLADVEKKPSTSEETQLDKSDKDKVEKSAIELIEGEEK
                                     1360       1370       1380       1390       1400       1410       1420       1430       1440

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HS -----IVPENAAHQGA---NRTSSLENHDQMSVNCRTL-----LSESPGKLHV-----EV-----SADLTAINGLS
MM -----VVPENAAHQGA---TR-PSLENHDQMSVNCRL-----PAESPAKLS-----EV-----SDLTAINGLS
RN -----VVPENAAHQGA---TR-PSLENHDQMSVNCRL-----PAESPAKLS-----EV-----SDLTAINGLS
XT -----TAPDASAASA---NSVIYMQINDQMSVNCPTP-----CQKSLSHLQTV-----CF-----SDDYKAINGIS
DR -----SDSNHTAPESC---S-----MEVSQPPGAPNT-----PDEKLETL-----TL-----PVTDL-NKD
CE IDDGEEFLNIGVWDPEEAVRIGVEISSRDRMDGEGDQTVGLTQGLHDGNIISDEDDLPFVMHDTYARL-----T-SNRNGIGAWS
DM LQEADDFFIEIGTWSNDMADDIASFN-----QEDDDE-----DDAFHLVPLPANVKFCDDQQTRYGSPTFWDVSNGESGFK
                                     1450       1460       1470       1480       1490       1500       1510       1520       1530

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HS YNQNLA-----NGSY-----D-----LANRDF-----CQGN-----
MM YKNKLA-----NGSY-----D-----LVNRDF-----CQGN-----
RN YNKSAL-----NGSY-----D-----LVNRDF-----CQGN-----
XT CN-GLT-----NGDW-----E-----AESAAC-----FQKD-----
DR CFPNLP-----NGTQADSDPD-----PHRSDV-----CQCS-----
CE GSESIV--PSGWDGWDGPPD-----NSMPMPQILGGPGGLNVQAL-----MADVGRVFPDSTASSLSQTVFESTVSPP
DM GPKSSQNKQGGKKAAGPAKPTFNYYDSNLSGSSYDDDDNAGPLNYMHNNYSSDDDDVADDIDAGRIAFTSKN-----EAETIETA
                                     1540       1550       1560       1570       1580       1590       1600       1610       1620

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Yfs (RCV000056333)



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HS -----QLNYFKQEI-----PVQPTTSYSIQ-----NLYSY-----ENQP-----
MM -----QLNYFKQEI-----PVQPTTSYPIQ-----NLNYN-----ENQP-----
RN -----QLTYFKQEI-----PVQPTTSYPIQ-----NLNYN-----ENQP-----
XT -----ERITCKQEI-----PEKS-TSFHVQ-----NLPK-----ENQP-----
DR -----QLGPLERDL-----STQTTTSVSVR-----PSPAG-----EPQP-----
CE KQLTKEEEQFKKLQNDL-----LKQAKERLEALE-MSEDMKPRRL--EDTV-----NLEDYGDQDE-----NQEDNTP-
DM QEVEKRQKQLSIIQATNANERYQQTKNLLIGFNFKHEDQKEPATIRYEESIACKLTEIESGGMLVPHDQQLVLKRSDAAEQVAKVSM
                                     1630       1640       1650       1660       1670       1680       1690       1700       1710

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HS -----QPSDE-----CT--LLSNKYLDGNAN-KS-----T-----SDGSPVMAMV-----PGTTD-----TI
MM -----KPSNE-----CP--LLSNKYLDGNAN-TS-----T-----SDGSPAVSTM-----PAMMN-----AV
RN -----TPSNE-----CP--LLSNKYLDGNAN-TS-----T-----SDGSPAGSPR-----PAMMT-----AV
XT -----L-LKE-----CT--LS--NSDGNVS-KP-----T-----SDECPSTC-----PAMMT-----AV
DR -----WPSDE-----CT--GRSSDLCPDHVK-KP-----T-----SKHCPKSETA-----TSTPA-----PS
CE ---TNFPKTIDEEIEELSIGARKKQEIIDNAA-KTDVLE-----RENCEVLPVA-----INEKSRFSFEKE-----SKAI
DM ELLKQLLPY-VNEDVLAKKLDRRELLLSDLVELNADWVARHEQETYNVNGCGDSFDNNDHRLNLDEKQLKQYERIEIEPPTSKAI
                                     1720       1730       1740       1750       1760       1770       1780       1790       1800

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Lfs (RCV000056332)



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HS QVLKGRMDSEQSPSIGYSSRTLGPNPGLILQALTLASNADGFLNLERLEMLGDSFLKHAITTYLFTCTYPDAHEGRLSYMRSSKKVSNCLNLYR
MM KALKDRMDSEQSPSIGYSSRTLGPNPGLILQALTLASNADGFLNLERLEMLGDSFLKHAITTYLFTCTYPDAHEGRLSYMRSSKKVSNCLNLYR
RN EALGERTDSEQSPSIGYSSRTLGPNPGLILQALTLASNADGFLNLERLEMLGDSFLKHAITTYLFTCTYPDAHEGRLSYMRSSKKVSNCLNLYR
XT -----TSDMHYDGLSNRHSSTLGPNPGLILQALTLASNADGFLNLERLEMLGDSFLKHAITTYLFTCTYPDAHEGRLSYMRSSKKVSNCLNLYR
DR ETSSEDCRSACAGPAWDSPTLGPNPGLILQALTLASNADGFLNLERLEMLGDSFLKHAITTYLFTCTYPDAHEGRLSYMRSSKKVSNCLNLYR
CE NGRILQRSEEEVYHSDSIDLGPNSPCLLLTALTITSNADGFLNLERLEMLGDSFLKHAITTYLFTCTYPDAHEGRLSYMRSSKKVSNCLNLYR
DM TSAIL-PAGFSFDRQPDVLGHPGPSIILQALTLASNADGFLNLERLEMLGDSFLKHAITTYLTYIYENVHEGKLSHLRSKQVANLNLYR
                                     1810       1820       1830       1840       1850       1860       1870       1880       1890

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HS LGKKKGLPSRMVYSIFDPPVNWLPFGYVYVNDKSNSEKWEKDEMTKDCMLANGKLDEYEEDEEESLWMWRAPKEEADEDDFLDYDQE
MM LGKKKGLPSRMVYSIFDPPVNWLPFGYVYVNDKSNSEKWEKDEMTKDCMLANGKLGEACEE-----EDLTWRAPKEEADEDDFLDYDQE
RN LGKKQGLPSRMVYSIFDPPVNWLPFGYVYVNDKSNSEKWEKDEMTKDCMLANGKLGEACEE-----EEELAWRAPKEEADEDDFLDYDQE
XT LGKKKGLPSRMVYSIFDPPVNWLPFGYVYVNDKSNSEKWEKDEMTKDCMLANGKLGEACEE-----EEELAWRAPKEEADEDDFLDYDQE
DR LGKKKGLPSRMVYSIFDPPVNWLPFGYVYVNDKSDTKWSDENK-----DLANGKASDDEDDDEDEPEAEARPEKEDVNVVEDFLYYE
CE LGKKLGIQLIVANKFPAHDSWLPCCYIPTCFKAPNTDDAEK-----DNEIE-----AKIPHTHHWKLADL-----LD-----
DM LGRRKRLGEYMIATKFDHDSWLPCCYIYVPEKEKALIE-----AKIPHTHHWKLADL-----LD-----
                                     1900       1910       1920       1930       1940       1950       1960       1970       1980

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HS HIRFIDNMLMGSGAFVKKISLSPFSTTDSAYEWMKPKKSSLSGMPFSSDFEDFDYSSWDAMCYLDPSKAVEEDDFVVGFWNPSEENCVD
MM HIQFIDNMLMGSGAFVKKISLSPFASDSAYEWMKPKKSLGSPFASGLDFDYSSWDAMCYLDPSKAVEEDDFVVGFWNPSEENCVD
RN HIQFIDNMLMGSGAFVKKISLSPFASDSAYEWMKPKKSLGSPFASGLDFDYSSWDAMCYLDPSKAVEEDDFVVGFWNPSEENCVD
XT HIRFIDNMLMGSGAFVKKISLSPFADPDQYEWAPKKPLLESSQPCDFDDFDYSSWDAMCYLDPSKAVEEDDFVVGFWNPSEENCVD
DR HIRFIDNMLMGSGAFVKKISLSPFADPDQYEWAPKKKAAHNSHFSDGGADEFDYSSWDAMCYLDPSKAVEEDDFVVGFWNPSEENCVD
CE ---RILDGQVIE-----IDPT-----NPNASLNFLLNWLQWQFQTFLEKIGYRFRKRAYLQAFTHASYINNRTGCVQRLEFLGDAVLDYM
DM -IKNLSSVQICEMV-----REKADALGLEQNGGAQN-----KTFNKGTWDIGGVSXST--TDGI-----ET-----IPFKQARVGN-
                                     1990       2000       2010       2020       2030       2040       2050       2060       2070

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L>R (RCV000004725)



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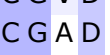
HS TGKQISYDLHTHEQCIAKSIADCEVALLGCYLTSCGERAAQLFLCSLGLKVLPIKRTSREKALCPDRENGSSQKLSVSCAAASV
MM TGKQISYDLHTHEQCIAKSIADCEVALLGCYLTSCGERAAQLFLCSLGLKVLPIKRTSREKALCPDRENGSSQKLSVSCAAASV--G
RN TGKQISYDLHTHEQCIAKSIADCEVALLGCYLTSCGERAAQLFLCSLGLKVLPIKRTSREKALCPDRENGSSQKLSVSCAAASV--S
XT AGKQISYDLHTHEQCIAKSIADCEVALLGCYLTSCGERAAQLFLCSLGLKVLPIKRTSREKALCPDRENGSSQKLSVSCAAASV--S
DR TGKQISYDLHTHEQCIAKSIADCEVALLGCYLTSCGERAAQLFLCSLGLKVLPIKRTSREKALCPDRENGSSQKLSVSCAAASV--S
CE DDISPLPNYLLTQQHISDKSIADCEVALLGVHLLTLPNLTGPNVLMVNMGLKVLQKQDSDVPSPLRRF-----KFNKAYLLQAFTHASYINNRTGCVQRLEFLGDAVLDYM
DM DFCSCIPYNLVQSHSIPDKSIADCEVALLGAYLIECGPRGALLFMAWLGVRLVITRQLDGGNQEQRIIPGSTKPN-----DVRV-
                                     2080       2090       2100       2110       2120       2130       2140       2150       2160

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D>Y (RCV000735853)



(RCV000735852)



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HS SRSSVLKDSSEYGLKIPPRCMFDHPDAD-KTLNHL--ISGFENFEKKINYRFKNKAYLLQAFTHASYHYNTITDCYQRLEFLGDAILDY
MM SRSSAGKDSSEYGLKIPPRCMFDHPDAD-KTLNHL--ISGFETFEKKINYRFKNKAYLLQAFTHASYHYNTITDCYQRLEFLGDAILDY
RN PRSSAGKDSSEYGLKIPPRCMFDHPDAD-KTLNHL--ISGFENFEKKINYRFKNKAYLLQAFTHASYHYNTITDCYQRLEFLGDAILDY
XT TDLSSCKGLDYGWLKIPPRCMFEHPDAE-KTLNHL--ISGFENFEKKINYRFKNKAYLLQAFTHASYHYNTITDCYQRLEFLGDAILDY
DR ---GGSAELQYGYWLKIPPRCMFEHPDAE-KTLNHL--ISGFENFEKKINYRFKNKAYLLQAFTHASYHYNTITDCYQRLEFLGDAILDY
CE -----IDPT-----NPNASLNFLLNWLQWQFQTFLEKIGYRFRKRAYLQAFTHASYINNRTGCVQRLEFLGDAVLDYM
DM ---AENVVTYGAWPTPRSPLLHFAPNATELLDQL--LSGFEEFEESLGYKFRDRSYLLQAFTHASYNNRITGCVQRLEFLGDAVLDY
                                     2170       2180       2190       2200       2210       2220       2230       2240       2250

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HS ITKHYLEDPRQHSPGVLTDLRSLVNNTIFASLAVKYDYHKYFKAVSPELFHVIDDVFQFQLEKNEMQGMDSL-----RRSEEDEEKED
MM ITKHYLEDPRQHSPGVLTDLRSLVNNTIFASLAVKYDYHKYFKAVSPELFHVIDDVFQFQLEKNEMQGMDSL-----RRSEEDEEKED
RN ITKHYLEDPRQHSPGVLTDLRSLVNNTIFASLAVKYDYHKYFKAVSPELFHVIDDVFQFQLEKNEMQGMDSL-----RRSEEDEEKED
XT ITKHYLEDPRQHSPGVLTDLRSLVNNTIFASLAVKYDYHKYFKAVSPELFHVIDDVFQFQLEKNEMQGMDSL-----RRSEEDEEKED
DR ITKHYLEDPRQHSPGVLTDLRSLVNNTIFASLAVKYDYHKYFKAVSPELFHVIDDVFQFQLEKNEMQGMDSL-----RRSEEDEEKED
CE ITRYLFEDSRQYSPGVLTDLRSLVNNTIFASLAVKFEFQKHFIAMCPGLYHMEIEKFKVLCSEARNFDTNFNAEMMYMTTEEIEDGQEE
DM ITRHYLEDPRQHSPGVLTDLRSLVNNTIFASLAVRHGFHKFHRHLSPLGNDVIDRVRIRIQENGHCII--SEEYULL--SEEECDAD
                                     2260       2270       2280       2290       2300       2310       2320       2330       2340

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S>X (RCV000023521)



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HS IEVPKAMGDIFESLAGAIYDMSGMSLETVVQVYVPMMPPLIEKFSANVRSPRVRELLMEPEATAKFSPAERTY-DGKVRVTVEVVGKGF
MM IEVPKAMGDIFESLAGAIYDMSGMSLETVVQVYVPMMPPLIEKFSANVRSPRVRELLMEPEATAKFSPAERTY-DGKVRVTVEVVGKGF
RN IEVPKAMGDIFESLAGAIYDMSGMSLETVVQVYVPMMPPLIEKFSANVRSPRVRELLMEPEATAKFSPAERTY-DGKVRVTVEVVGKGF
XT IEVPKAMGDIFESLAGAIYDMSGMSLETVVQVYVPMMPPLIEKFSANVRSPRVRELLMEPEATAKFSPAERTY-DGKVRVTVEVVGKGF
DR IEVPKAMGDIFESLAGAIYDMSGMSLETVVQVYVPMMPPLIEKFSANVRSPRVRELLMEPEATAKFSPAERTY-DGKVRVTVEVVGKGF
CE IEVPKAMGDIFESLAGAIYDMSGMSLETVVQVYVPMMPPLIEKFSANVRSPRVRELLMEPEATAKFSPAERTY-DGKVRVTVEVVGKGF
DM VEVPKALGDQVFESIAGAIYDMSGMSLETVVQVYVPMMPPLIEKFSANVRSPRVRELLMEPEATAKFSPAERTY-DGKVRVTVEVVGKGF
                                     2350       2360       2370       2380       2390       2400       2410       2420       2430

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HS KGV
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